Scalability & Future Plan

Date	5 July 2026
Team ID	XXXXXX
Project Name	**Emotion Detection and Learning Support Engine**
Maximum Marks	1 Mark

Scalability & Future Plan

This document explains how the Emotion Detection and Learning Support Engine (EDLSE) can be expanded to support a larger number of users, improve performance, strengthen security, and incorporate advanced Artificial Intelligence capabilities in future releases.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Supports limited concurrent users in the current Streamlit deployment.	Reduced performance when multiple users access the application simultaneously.	High
2	Uses local SQLite/JSON storage.	Limited scalability and centralized data management.	High
3	Emotion detection accuracy may vary with poor lighting, noisy audio, or low-quality input.	May affect prediction accuracy and user experience.	Medium
4	Supports a limited set of emotion categories.	Cannot recognize complex or mixed emotional states.	Medium

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Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance Streamlit application.	Deploy on cloud infrastructure with multi-user support, auto-scaling, and load balancing.
2	Data Storage	SQLite / JSON files.	Migrate to PostgreSQL, Firebase, or MongoDB for centralized and scalable

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
			storage.
3	Performance	CPU-based AI inference.	Optimize deep learning models, enable GPU acceleration, and apply model optimization techniques.
4	Security	Basic authentication.	Implement secure authentication, encrypted data storage, HTTPS, and role-based access control.
5	AI Capabilities	Basic facial, voice, and text emotion detection.	Integrate multimodal emotion recognition, explainable AI, and adaptive learning models.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud Deployment using Streamlit Community Cloud	1 Month	Enable online accessibility and remote access for users.
Phase 2	Database Integration (PostgreSQL/Firebase)	2 Months	Improve scalability, reliability, and centralized data management.
Phase 3	Advanced AI-Based Multimodal Emotion Detection and Personalized Learning	4 Months	Increase emotion recognition accuracy and provide more effective learning recommendations.
Phase 4	Mobile Application, Multi-language Support, and AI Chatbot Integration	6 Months	Enhance accessibility, improve user engagement, and support a wider range of learners.